

AISHWARYA MANDYA YOGANANDA

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EDUCATION

Binghamton University, State University of New York
Master of Science in Computer Science

Aug 2024 – May 2026

Visvesvaraya Technological University
Bachelor of Engineering in Information Science and Engineering

Sep 2020 – May 2024

EXPERIENCE

Machine Learning Software Engineer | *Manifoldz, NY, USA*

May 2026 – Present

- Engineered the **MSCTransToReportTool** by implementing core logic modules, validating transformation workflows, and documenting **100% of cell-level code logic**, enabling Samar's team to adopt the tool and reducing onboarding time by **40%**.
- Improved **ML/data pipeline reliability** by identifying, categorizing, and logging **15+ software defects and data inconsistencies** into an incident tracking system, ensuring **100% issue traceability** and accelerating structured triage across the MSC pipeline.
- Led cross-functional **QA/QC validation** with Tripp across the **MSC Report Tool**, analyzing output accuracy, model-ready data quality, surfacing **20+ reporting errors**, and creating a **5-step remediation roadmap** that reduced downstream rework risk by **30%**.

Machine Learning Software Engineering Intern | *Manifoldz, NY, USA*

Sep 2025 – Apr 2026

- Addressed high ASR error rates in legal audio by fine-tuning **Whisper** with **Hugging Face Transformers**, implementing standardized preprocessing, data augmentation, and optimization techniques to reduce **ASR error rate by 43.3%** in production transcription pipelines.
- Improved slow routing inference workflows by building a **real-time ML pipeline** ingesting live **NWS** weather data, processing conditions through API-driven services, and deploying **LLM-driven route optimization** with OpenAI APIs to decrease inference latency by **30%**.
- Enhanced model and alert performance by retraining **TensorFlow deep learning** and **Hugging Face NLP models** on curated domain data, applying feature engineering, validation, and evaluation pipelines to improve **accuracy by 18%** and alert reliability by **20%**.
- Built a **0→1 production ML-enabled backend system** for a mobile application by using **Laravel** to integrate model inference workflows, REST APIs, and scalable backend architecture, enabling real-time task orchestration and seamless cross-platform user interactions.

Research Assistant | *Binghamton University, Binghamton, NY*

Jan 2025 – Apr 2025

- Built a **secure real-time ML data pipeline** on FPGA using **C, RTOS, and AES-128 encryption**, enabling sensor data transmission, validation, and preprocessing for downstream analytics while achieving **99.9% data reliability** in constrained environments.
- Strengthened **high-throughput streaming data pipelines** through profiling, performance tuning, and system-level testing, reducing latency by **40%**, sustaining **100 FPS throughput** and **95% uptime**, and improving data readiness for real-time ML inference workflows.

Data Scientist Intern | *BrainOVision Solutions, Bangalore, India*

Apr 2022 – May 2022

- Redesigned **ML data processing workflows** using **Python, Pandas, NumPy, and SQL**, automating preprocessing, feature validation, quality checks, and removing manual bottlenecks to improve pipeline efficiency by **30%** and support scalable model-ready data preparation.
- Developed a scalable **NLP-based sentiment analysis pipeline** using **NLTK and Python**, automating text classification, feature extraction, model scoring, and BI dashboard integration to reduce manual analysis time by **50%** and enable real-time business insights.

PROJECTS

AI GTM Copilot | *Python, OpenAI API, SQLite, Pandas*

- Built an **AI-driven GTM assistant** using **LLMs, OpenAI API, and MCP-based workflows**, consolidating **5+ tools** into a unified pipeline with prompt orchestration to improve lead qualification efficiency by **40%** and automate account research and outreach generation.
- Optimized **inference and lead-processing pipelines** using **Python, Pandas, and SQLite**, enabling scalable enrichment and personalized outreach generation for **1K+ leads/day** while improving response consistency, workflow speed, and operational efficiency.

Audit Readiness AI | *Python, OpenAI API, NLP*

- Built an **LLM-powered compliance assistant** using **RAG-based pipelines** to ingest, retrieve, and analyze enterprise policy documents, automating regulatory gap analysis, reducing manual audit effort by **60%**, and improving compliance recommendation accuracy at scale.
- Optimized **retrieval and generation workflows** using **OpenAI APIs**, prompt engineering, chunking strategies, vector search, and context-aware remediation logic to process **10K+ documents**, generate policy gap outputs, and improve response consistency by **25%**.

AI Incident Investigation Agent | *LangGraph, RAG, FastAPI, OpenAI, AWS*

- Built a **scalable LLM investigation pipeline** using **LangGraph, RAG, FastAPI, PostgreSQL, and Redis caching**, enabling semantic retrieval, context-aware generation, and workflows to reduce investigation time by **40%** and improve response accuracy by **30%**.
- Enhanced **retrieval and inference workflows** on **AWS** by implementing prompt orchestration, caching strategies, scalable API serving, and latency optimization, supporting **100+ queries/day** with sub-**2s** responses while improving system reliability and output consistency.

TECHNICAL SKILLS

Programming Languages & Frameworks: Python, C, C++, Java, JavaScript, PHP, React, Node.js, Laravel, Flask

Data & Analytics: SQL, PostgreSQL, SQLite, MongoDB, Pandas, Matplotlib, Tableau, Power BI, Spark, PySpark, Snowflake

ML/AI: Scikit-learn, TensorFlow, PyTorch, Hugging Face, Whisper, OpenCV, NLP, Keras, Feature Engineering, Model Fine-tuning

Agentic & LLM Tools: Claude API, Anthropic SDK, OpenAI API, LangChain, MCP, RAG, Prompt Engineering

Cloud & DevOps: Docker, Kubernetes, AWS (EC2, IAM, S3), GCP (Compute Engine), CI/CD Pipelines